

# (Applications Development and Emerging Technologies)

## TECHNICAL-SUMMATIVE ASSESSMENT

# 1

### PHP OUTPUT, VARIABLE FAMILIARIZATION, OPERATORS AND CONTROL STRUCTURE

|                                   |                    |             |
|-----------------------------------|--------------------|-------------|
| <b>Student Name / Group Name:</b> | Kirk Marcus Garing |             |
| <b>Members (if Group):</b>        | <b>Name</b>        | <b>Role</b> |
|                                   |                    |             |
|                                   |                    |             |
|                                   |                    |             |
| <b>Section:</b>                   | TC27               |             |
| <b>Professor:</b>                 | Sir. Calleja       |             |

#### **I. PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE**

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements.

#### **II. COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE**

- Understand and apply best practices and standards in the development of website.

#### **III. INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE**

At the end of this exercise, students must be able to:

- Familiarize various Web Architecture, tools that used in PHP
- The basic understanding before using PHP
- Familiarize in environment of web developing
- Use of comments, variables and Echo / Print
- To understand the different types of operators that are available on PHP.
- To know what is operator precedence and operator associativity in PHP.
- To use escape sequence properly in the program.
- To know the different approach of control structures.
- To know the fundamentals syntax for conditional and looping structures.
- To properly use the compound expression using the logical operators.
- To know the rules of break, continue, and goto statements.

#### **IV. BACKGROUND INFORMATION**

## Running PHP Scripts

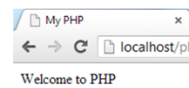
```
<!-- welcome.php -->
<html>
<head>
<title>My PHP</title>
</head>
<body>

<?php
    echo 'Welcome to PHP';
?>

</body>
</html>
```

All php scripts must enclosed with  
<?php ... ?> (standard tag)  
<? ... ?> (short open tag)  
<script language="php">  
....  
</script> (script tag)  
<% ... %> (asp style tag)

OUTPUT:



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## An introduction to PHP web programming

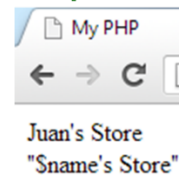
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## Using ‘ ‘ (pair single quote), “ “ (pair double quotes) and . (dot character) PHP

Example:

```
<?php
$name = "Juan";
echo "$name's Store". "<br/>";
echo "'$name\'s Store' ". "<br/>";
//statements below will produced errors
//echo "displaying double quote " ";
//echo "dispalying single quote ' ";
```

Output:



## PHP Operators and Control Structures

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### Arithmetic Operator

| Example                | Label          | Outcome                         |
|------------------------|----------------|---------------------------------|
| <code>\$a + \$b</code> | Addition       | Sum of \$a and \$b              |
| <code>\$a - \$b</code> | Subtraction    | Difference of \$a and \$b       |
| <code>\$a * \$b</code> | Multiplication | Product of \$a and \$b          |
| <code>\$a / \$b</code> | Division       | Quotient of \$a and \$b         |
| <code>\$a % \$b</code> | Modulus        | Remainder of \$a divided by \$b |

*Beginning PHP and MySQL 3<sup>rd</sup> Edition by: W. Jason Gilmore pp.124*

### Assignment Operator

| Example               | Label                     | Outcome                            |
|-----------------------|---------------------------|------------------------------------|
| <code>\$a = 5</code>  | Assignment                | \$a equals 5                       |
| <code>\$a += 5</code> | Addition-assignment       | \$a equals \$a plus 5              |
| <code>\$a *= 5</code> | Multiplication-assignment | \$a equals \$a multiplied by 5     |
| <code>\$a /= 5</code> | Division-assignment       | \$a equals \$a divided by 5        |
| <code>\$a .= 5</code> | Concatenation-assignment  | \$a equals \$a concatenated with 5 |

*Beginning PHP and MySQL 3<sup>rd</sup> Edition by: W. Jason Gilmore pp.126*

## PHP Operators and Control Structures

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### PHP Control Structures: Conditional Statements

#### if statement

syntax:  
`if(expression) {  
 statement...  
}`

#### else statement

syntax:  
`if(expression) {  
 statement...  
} else {  
 statement  
}`

#### elseif statement

syntax:  
`if(expression) {  
 statement...  
}  
elseif(expression) {  
 statement...  
} else {  
 statement...  
}`

**while statement**

syntax:

```
while (expression) {  
    statement...  
}
```

**for statement**

syntax:

```
for (expr1 ; expr2 ; expr3)  
{  
    statement...  
}
```

**do ... while statement**

syntax:

```
do {  
    statement...  
} while (expression) ;
```

**V. GRADING SYSTEM / RUBRIC (please see separate sheet)****VI. LABORATORY ACTIVITY**

1. The student will create a student registration form using HTML and CSS with the integration of PHP Scripts please refer to the attached image for the example.
  - a. All user entries from the student registration form will be converted in variables
  - b. In the output, they need to call for the declared variables and do some string formats like name cases and numbers

## Student Registration Form

|                                                |  |                       |  |                                                                                                                       |  |
|------------------------------------------------|--|-----------------------|--|-----------------------------------------------------------------------------------------------------------------------|--|
| For School Use - Permission to Register: _____ |  | Date: _____           |  | <input type="checkbox"/> ESL <input type="checkbox"/> SPED <input type="checkbox"/> IPRC <input type="checkbox"/> ISA |  |
| <small>(Administrator's Initials)</small>      |  |                       |  |                                                                                                                       |  |
| Student Number: _____                          |  | Entry Date: _____     |  | Entry Type: _____ Grade: _____                                                                                        |  |
| OEN: _____                                     |  | Class/Homeform: _____ |  |                                                                                                                       |  |

Is the student currently expelled from any school or school board? ☐ Yes ☐ No

**Student Information**

Full Legal Name: \_\_\_\_\_

Last Name                      First Name                      Middle Name

Preferred Name: \_\_\_\_\_

(if different)                      Last Name                      First Name                      Middle Name

Gender: ☐ Male ☐ Female      Date of Birth: \_\_\_\_\_

Year                      Month                      Day

If the student has other siblings in this school, please list them:

\_\_\_\_\_

For School Use - Proof of Birth: ☐ Copy in OSR   ☐ Baptismal Record   ☐ Birth Registration   ☐ Birth Certificate   ☐ Immigration Papers/Card   ☐ Passport   ☐ Other

**Previous School Information**

Name of Previous School Board / Municipality: \_\_\_\_\_ Last date attended: \_\_\_\_\_

Name of Previous School: \_\_\_\_\_ Grade at previous school: \_\_\_\_\_

Language of Instruction: ☐ English   ☐ French   ☐ Other      Reason for Transfer: \_\_\_\_\_

Did the student ever attend a Waterloo Region District School Board school in the past? ☐ Yes ☐ No    If yes, name school(s):

\_\_\_\_\_

**Health Information**

Medical Conditions (include information on special equipment or medication, if required):

\_\_\_\_\_

\_\_\_\_\_

**For School Use**

Immunization Record provided: ☐ Yes ☐ No

Does the student require an epi-pen? ☐ Yes ☐ No

**Citizenship Information**

Birth Country: \_\_\_\_\_ If Canada, Province of Birth: \_\_\_\_\_

Country of Citizenship: \_\_\_\_\_

If student not born in Canada, provide date student entered Canada to live for the first time: \_\_\_\_\_

Year                      Month                      Day

**For School Use - Status in Canada:**

☐ Canadian Citizen      ☐ Permanent Resident/ Landed Immigrant  
☐ Study Permit/ Student Visa      ☐ Refugee Claimant  
☐ Native Ancestry  
☐ Other (specify): \_\_\_\_\_  
☐ Verification in OSR

localhost/PHPTSA1/

## Student Registration

### PERSONAL INFORMATION

Full Name  
Enter full name

Email Address  
example@school.edu

Gender  
Male

Civil Status  
☐ Single ☐ Married

Date of Birth  
mm/dd/yyyy

### ACADEMIC INFORMATION

Student Number

Grade Level  
Grade 7

localhost/PHPTSA1/

ACADEMIC INFORMATION

Student Number

Grade Level

Grade 7

Section

e.g. ICT-1

Scholarship Status

None

EDUCATIONAL BACKGROUND

Grade School

High School

Senior High School

EMERGENCY CONTACT

Contact Name



## EMERGENCY CONTACT

Contact Name

Relationship

Contact Number

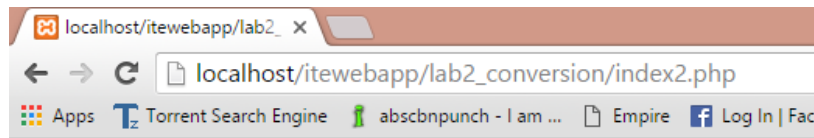
## ADDRESS

City

Country

Submit Registration

2. Create a php program that will display the same output below. Use control structures to display the multiplication table with alternating color.



# Multiplication Table

|   |    |    |    |    |    |    |    |    |    |     |
|---|----|----|----|----|----|----|----|----|----|-----|
| 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10  |
| 0 | 2  | 4  | 6  | 8  | 10 | 12 | 14 | 16 | 18 | 20  |
| 0 | 3  | 6  | 9  | 12 | 15 | 18 | 21 | 24 | 27 | 30  |
| 0 | 4  | 8  | 12 | 16 | 20 | 24 | 28 | 32 | 36 | 40  |
| 0 | 5  | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50  |
| 0 | 6  | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 54 | 60  |
| 0 | 7  | 14 | 21 | 28 | 35 | 42 | 49 | 56 | 63 | 70  |
| 0 | 8  | 16 | 24 | 32 | 40 | 48 | 56 | 64 | 72 | 80  |
| 0 | 9  | 18 | 27 | 36 | 45 | 54 | 63 | 72 | 81 | 90  |
| 0 | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |

Multiplication Table

|   |    |    |    |    |    |    |    |    |    |     |
|---|----|----|----|----|----|----|----|----|----|-----|
| 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10  |
| 0 | 2  | 4  | 6  | 8  | 10 | 12 | 14 | 16 | 18 | 20  |
| 0 | 3  | 6  | 9  | 12 | 15 | 18 | 21 | 24 | 27 | 30  |
| 0 | 4  | 8  | 12 | 16 | 20 | 24 | 28 | 32 | 36 | 40  |
| 0 | 5  | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50  |
| 0 | 6  | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 54 | 60  |
| 0 | 7  | 14 | 21 | 28 | 35 | 42 | 49 | 56 | 63 | 70  |
| 0 | 8  | 16 | 24 | 32 | 40 | 48 | 56 | 64 | 72 | 80  |
| 0 | 9  | 18 | 27 | 36 | 45 | 54 | 63 | 72 | 81 | 90  |
| 0 | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |

## VII. QUESTION AND ANSWER

1. What is a variable?

**A variable stores data like numbers, text, or values in a PHP program. It works as a container you can change anytime while the program runs.**

2. What are the rules in creating a variable?

**A variable starts with a dollar sign. It must begin with a letter or underscore. It cannot start with a number. It should not contain space. It is case sensitive.**

3. Is it important to know HTML and CSS before using PHP? Explain

**Yes. HTML and CSS help you build and design the webpage, while PHP handles the logic and data. PHP works inside HTML, so knowing both helps you create full websites.**

4. What is the difference between operator precedence and operator associativity?

**Operator precedence decides which operation runs first in an expression. Operator associativity decides the order when operators have the same priority, like left to right or right to left.**

5. What are the different control structures? Explain each

Control structures guide how a program runs.

- Sequence runs code step by step.
- Selection chooses between options using if, else, or switch.
- Iteration repeats code using loops like for, while, or do while.

6. Explain the rules of break, continue, and goto statements.

break stops a loop or switch immediately. continue skips the current loop cycle and moves to the next one. goto jumps to a labeled part of the code, but it is rarely used because it makes code harder to read.

## VIII. REFERENCES

1. [https://www.w3schools.com/php/func\\_string\\_echo.asp](https://www.w3schools.com/php/func_string_echo.asp)
2. <https://www.w3schools.com/css/>
3. <https://www.w3schools.com/html/>
4. [https://www.w3schools.com/php/php\\_variables.asp](https://www.w3schools.com/php/php_variables.asp)
5. <https://www.w3resource.com/php/operators/arithmetic-operators.php>
6. [https://www.tutorialspoint.com/php/php\\_arithmetic\\_operators\\_examples.htm](https://www.tutorialspoint.com/php/php_arithmetic_operators_examples.htm)
7. [https://www.w3schools.com/php/php\\_if\\_else.asp](https://www.w3schools.com/php/php_if_else.asp)
8. [https://www.w3schools.com/php/php\\_switch.asp](https://www.w3schools.com/php/php_switch.asp)
9. [https://www.w3schools.com/php/php\\_looping.asp](https://www.w3schools.com/php/php_looping.asp)
10. [https://www.w3schools.com/php/php\\_looping\\_while.asp](https://www.w3schools.com/php/php_looping_while.asp)
11. [https://www.w3schools.com/php/php\\_looping\\_do\\_while.asp](https://www.w3schools.com/php/php_looping_do_while.asp)
12. [https://www.w3schools.com/php/php\\_looping\\_for.asp](https://www.w3schools.com/php/php_looping_for.asp)
13. [https://www.w3schools.com/php/php\\_looping\\_foreach.asp](https://www.w3schools.com/php/php_looping_foreach.asp)
14. [https://www.w3schools.com/php/php\\_looping\\_break.asp](https://www.w3schools.com/php/php_looping_break.asp)

**Note: The following rubrics/metrics will be used to grade students' output.**

| <b>Program (100 pts.)</b>        | <b>(Excellent)</b>                                                            | <b>(Good)</b>                                                                                       | <b>(Fair)</b>                                                                                           | <b>(Poor)</b>                               |
|----------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Program execution (20pts)</b> | Program executes correctly with no syntax or runtime errors <b>(18-20pts)</b> | Program executes with less than 3 errors <b>(15-17pts)</b>                                          | Program executes with more than 3 errors <b>(12-14pts)</b>                                              | Program does not execute <b>(10-11pts)</b>  |
| <b>Correct output (20pts)</b>    | Program displays correct output with no errors <b>(18-20pts)</b>              | Output has minor errors <b>(15-17pts)</b>                                                           | Output has multiple errors <b>(12-14pts)</b>                                                            | Output is incorrect <b>(10-11pts)</b>       |
| <b>Design of output (10pts)</b>  | Program displays more than expected <b>(10pts)</b>                            | Program displays minimally expected output <b>(8-9pts)</b>                                          | Program does not display the required output <b>(6-7pts)</b>                                            | Output is poorly designed <b>(5pts)</b>     |
| <b>Design of logic (20pts)</b>   | Program is logically well designed <b>(18-20pts)</b>                          | Program has slight logic errors that do no significantly affect the results <b>(15-17pts)</b>       | Program has significant logic errors <b>(3-5pts)</b>                                                    | Program is incorrect <b>(10-11pts)</b>      |
| <b>Standards (20pts)</b>         | Program code is stylistically well designed <b>(18-20pts)</b>                 | Few inappropriate design choices (i.e. poor variable names, improper indentation) <b>(15-17pts)</b> | Several inappropriate design choices (i.e. poor variable names, improper indentation) <b>(12-14pts)</b> | Program is poorly written <b>(10-11pts)</b> |

|                             |                                                      |                                                                        |                                                                        |                                                                                   |
|-----------------------------|------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Delivery<br/>(10pts)</b> | The program was delivered on time.<br><b>(10pts)</b> | The program was delivered a day after the deadline.<br><b>(8-9pts)</b> | The program was delivered two days after the deadline. <b>(6-7pts)</b> | The program was delivered more than two days after the deadline.<br><b>(5pts)</b> |
|-----------------------------|------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------|